

MATEO ZITELLA MINI

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EXPERIENCE

Schleifring North America | Lowell, MA

August 2025 - Present

Manufacturing Engineer Coop

- Create Solid Edge models, technical drawings, and SLA3D print new and existing assembly fixtures
- Evaluate faulty slip rings using a microscope and multimeter to assign appropriate repair and quality test tasks to people in assembly and quality control through SAP
- Collaborated with the machine shop to create clear technical drawings and easily manufacturable designs
- Write and improve existing work instructions for assembly and rework of slip rings
- Update the format and contents of a company-wide fixture CAD tracking database in Excel

Northeastern DeLTA Lab | Burlington, MA

October 2024 - July 2025

Research Assistant

- Designed equipment for experiments involving X-ray optical testing of different material properties
- Referenced technical drawings to design adjustable fixtures for lab equipment that interface with the optical table
- Designed a safety cover for laser experiments in SolidWorks using 80/20 bars and acetal sheets

Northeastern Silicon Synapse Lab | Boston, MA

October 2024 - May 2025

Research Assistant for Aerobat Project

- Designed and prototyped a print-in-place design to increase ease of assembly for Aerobat, a flapping wing robot
- Planned toolpaths in Fusion360 to CNC machine lightweight aluminum gears for Aerobat
- Assembled and soldered electrical components to minimize the weight of wires

PROJECTS

Combat Robotics Club

Stormbringer - 3 Pound Combat Robot

January 2025 - June 2025

- Researched successful combat robot designs and compared different ideas for the robot's weapon system
- Investigated compatibility of electronics including a weapon motor, weapon ESC, drive ESC, and LiPo battery
- Designed the body and weapon, taking mechanical hardware, materials, and weight optimization into account

GALINT - 1 Pound Combat Robot

September 2024 - March 2025

- Designed iterations of the robot in SolidWorks to optimize performance while meeting the weight requirement
- 3D printed, soldered, and assembled the robot's frame and electronics
- Used hand drills and a dremel to make changes to the robot's frame and weapon after being 3D printed

Forge Product Development Lab

Smart Step

September - December 2024

- Developed a telescopic actuator mechanism using Solidworks for a height-changing smart cane, designed the body tube linkage system, and troubleshooted electronics and C++ code
- Collaborated with a team of 7 to select lightweight and cost efficient hardware
- Created an Onshape assembly and edited team member's designs to ensure integration into the overall design

NASA Student Launch Challenge

Subscale Rocket

September - November 2024

- Collaborated with a team of 7 people to design fin mounts for a 4.85' subscale rocket in SolidWorks and Onshape
- Designed and simulated a subscale rocket model using OpenRocket to analyze flight performance and stability

EDUCATION

Northeastern University College of Engineering | Boston, MA

May 2027

Candidate for Bachelor of Science in Mechanical Engineering

GPA: 3.57 (Dean's List)

- *Relevant Courses:* Fluid Mechanics, Dynamics, Intro to Material Science, Mechanics of Materials, Thermodynamics, Statics, Differential Equations & Linear Algebra, Physics 1-2, Calculus 1-3

SKILLS

Software: SolidWorks, Solid Edge, Onshape, Fusion360 CAM, SAP, ECTR, Adobe Illustrator, OpenRocket, Microsoft Office, Google Suite

Fabrication: Arduino, Soldering, SLA and FDM 3D printing, Laser cutting, CNC mill, Woodworking, Hand tools

Languages: Spanish, C++, Python, CSS, HTML, JavaScript, MATLAB

Interests: Art, Guitar, Anime, Product Development